ePGD: Statistical Foundations of Machine Learning

Midsem - Solutions

Total Marks: 30

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1. Two fair six-sided dice are rolled and the outcomes are recorded as pairs (i, j). Consider the events

$$A = \{(i,j) : i + j = 8\} \qquad \text{and} \qquad B = \{(i,j) : i > 3\}.$$

   (a) (2 marks) Find $P(A \cup B)$.

   Solution. There are 36 equally likely outcomes. We have

$$\begin{aligned} A =& \{(2,6),(3,5),(4,4),(5,3),(6,2)\} \\ B =& \{(4,1),(4,2),(4,3),(4,4),(4,5),(4,6) \\ & (5,1),(5,2),(5,3),(5,4),(5,5),(5,6) \\ & (6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\} \end{aligned}$$

$$|A| = 5, \qquad |B| = 18.$$

   The outcomes in both events are $(4,4), (5,3)$, and $(6,2)$, so $|A \cap B| = 3$. Hence

$$P(A \cup B) = \frac{|A| + |B| - |A \cap B|}{36} = \frac{5 + 18 - 3}{36} = \frac{5}{9}$$

   (b) (2 marks) Are A, B independent events?

   Solution.

$$P(A) = \frac{5}{36}, \qquad P(B) = \frac{1}{2}, \qquad P(A \cap B) = \frac{3}{36} = \frac{1}{12}.$$

   If A and B were independent, we would have

$$P(A \cap B) = P(A)P(B) = \frac{5}{36} \cdot \frac{1}{2} = \frac{5}{72}.$$

   Since $\frac{1}{12} \neq \frac{5}{72}$, the events are not independent.

2. A dataset $\mathcal{D}$ contains n distinct data points. A data scientist samples k data points independently with replacement, with each data point selected uniformly at random from $\mathcal{D}$. Let Φ_1 denote the number of distinct data points that appear exactly once among the k sampled data points.

   (a) Suppose $k = 3$ and $n = 3$.

      i. (2 marks) Find the probability mass function (pmf) of Φ_1.

      Solution. There are $3^3 = 27$ equally likely ordered samples.

      - $\Phi_1 = 0$: all three observations are equal. There are 3 such samples:

$$P(\Phi_1 = 0) = \frac{3}{27} = \frac{1}{9}.$$

      - $\Phi_1 = 1$: one value appears twice and another appears once. Choose the repeated value in 3 ways, the singleton in 2 ways, and its position in 3 ways:

$$P(\Phi_1 = 1) = \frac{3 \cdot 2 \cdot 3}{27} = \frac{2}{3}.$$

- $\Phi_1 = 3$: all three observations are distinct. There are $3! = 6$ samples:

$$P(\Phi_1 = 3) = \frac{6}{27} = \frac{2}{9}.$$

Thus

$$P(\Phi_1 = r) = \begin{cases} \frac{1}{9}, & r = 0, \\ \frac{2}{3}, & r = 1, \\ \frac{2}{9}, & r = 3, \\ 0, & \text{otherwise.} \end{cases}$$

ii. (1 mark) Compute $E[\Phi_1]$.
Solution.

$$E[\Phi_1] = 0 \cdot \frac{1}{9} + 1 \cdot \frac{2}{3} + 3 \cdot \frac{2}{9} = \frac{4}{3}$$

(b) (2 marks) Find $E[\Phi_1]$ for general n and k.
Solution. For each data point $r \in \mathcal{D}$, define

$$I_r = \begin{cases} 1, & \text{if } r \text{ appears exactly once,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\Phi_1 = \sum_{r \in \mathcal{D}} I_r.$$

A fixed data point appears exactly once if it is selected in one of the k positions and is not selected in the other $k - 1$ positions. Therefore

$$P(I_r = 1) = k \left(\frac{1}{n}\right) \left(1 - \frac{1}{n}\right)^{k-1}.$$

By linearity of expectation,

$$E[\Phi_1] = \sum_{r \in \mathcal{D}} E[I_r] = n\, k \left(\frac{1}{n}\right) \left(1 - \frac{1}{n}\right)^{k-1} = k \left(1 - \frac{1}{n}\right)^{k-1}$$

3. A data scientist studies the relationship between the number of product recommendations clicked by a customer during a browsing session and the number of items added to the shopping cart during that session. Let

- X denote the number of product recommendations clicked, and
- Y denote the number of items added to the shopping cart.

Suppose the joint probability mass function (PMF) of X and Y is given by:

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	0.08	0.12	0.04
$X = 1$	0.06	0.14	0.16
$X = 2$	0.02	0.18	0.20

Answer the following questions.

(a) (1 mark) Find the marginal PMF of X.

Solution. Sum each row:

$$p_X(x) = P(X = x) = \begin{cases} 0.24, & x = 0, \\ 0.36, & x = 1, \\ 0.40, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

(b) (1 mark) Find the marginal PMF of Y.

Solution. Sum each column:

$$p_Y(y) = P(Y = y) = \begin{cases} 0.16, & y = 0, \\ 0.44, & y = 1, \\ 0.40, & y = 2, \\ 0, & \text{otherwise.} \end{cases}$$

(c) (2 marks) Find the conditional PMF of X given $Y = 1$.

Solution. Since $P(Y = 1) = 0.44$,

$$P(X = x \mid Y = 1) = \frac{P(X = x, Y = 1)}{0.44}.$$

Hence

$$p_{X|Y}(x \mid 1) = P(X = x \mid Y = 1) = \begin{cases} 0.273, & x = 0, \\ 0.318, & x = 1, \\ 0.409, & x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

(leaving as a fraction is also accepted)

(d) (1 mark) Compute $P(Y \geq X)$.

Solution. The admissible pairs are

$$(0,0), (0,1), (0,2), (1,1), (1,2), (2,2).$$

Therefore

$$\begin{aligned} P(Y \geq X) &= 0.08 + 0.12 + 0.04 + 0.14 + 0.16 + 0.20 \\ &= 0.74 \end{aligned}$$

(e) (5 marks) Compute the correlation coefficient between X and Y,

$$\rho_{X,Y} = \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X) \operatorname{Var}(Y)}}.$$

Solution. From the marginal PMFs,

$$E[X] = \sum_x x\, P(X = x) = 0(0.24) + 1(0.36) + 2(0.40) = 1.16,$$

$$E[Y] = \sum_y y\, P(Y = y) = 0(0.16) + 1(0.44) + 2(0.40) = 1.24.$$

Also,

$$E[X^2] = \sum_x x^2\, P(X = x) = 0^2(0.24) + 1^2(0.36) + 2^2(0.40) = 1.96,$$

$$E[Y^2] = \sum_y y^2\, P(Y = y) = 0^2(0.16) + 1^2(0.44) + 2^2(0.40) = 2.04.$$

Thus

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = 1.96 - (1.16)^2 = 0.6144,$$
$$\mathrm{Var}(Y) = E[Y^2] - (E[Y])^2 = 2.04 - (1.24)^2 = 0.5024.$$

Further,

$$\begin{aligned} E[XY] &= \sum_{x,y} xy\, P(X = x, Y = y) \\ &= 0.14 + 2(0.16) + 2(0.18) + 4(0.20) \\ &= 1.62. \end{aligned}$$

Hence

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = 1.62 - (1.16)(1.24) = 0.1816.$$

Therefore

$$\begin{aligned} \rho_{X,Y} &= \frac{0.1816}{\sqrt{(0.6144)(0.5024)}} \\ &\approx 0.327 \end{aligned}$$

(leaving as a fraction is also accepted)

4. Let X and Y be jointly continuous random variables with joint probability density function

$$f_{X,Y}(x, y) = \begin{cases} x + y, & 0 \le x \le 1,\ 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) (2 marks) Verify that $f_{X,Y}$ is a valid joint probability density function.
Solution. On its support, $x + y \ge 0$, so $f_{X,Y}(x, y) \ge 0$. Moreover,

$$\begin{aligned} \int_0^1 \int_0^1 (x + y)\, dy\, dx &= \int_0^1 \left(x + \frac{1}{2} \right) dx \\ &= \frac{1}{2} + \frac{1}{2} = 1. \end{aligned}$$

Hence $f_{X,Y}$ is a valid joint PDF.

(b) (3 marks) Find the marginal probability density functions $f_X(x)$ and $f_Y(y)$.
Solution. For $0 \le x \le 1$,

$$\begin{aligned} f_X(x) &= \int_0^1 (x + y)\, dy \\ &= x + \frac{1}{2}. \end{aligned}$$

Thus

$$f_X(x) = \begin{cases} x + \dfrac{1}{2}, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Similarly,

$$f_Y(y) = \begin{cases} y + \dfrac{1}{2}, & 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(c) (2 marks) Find the joint cumulative distribution function

$$F_{X,Y}(x,y) = P(X \le x, Y \le y),$$

for $0 \le x \le 1$, $0 \le y \le 1$.

Solution. For $0 \le x, y \le 1$,

$$\begin{aligned} F_{X,Y}(x,y) &= \int_0^x \int_0^y (u+v)\, dv\, du \\ &= \int_0^x \left(uy + \frac{y^2}{2} \right) du \\ &= \frac{xy^2 + x^2 y}{2} \end{aligned}$$

(d) (2 marks) Using the joint CDF or the joint PDF, compute

$$P\left(\frac{1}{2} < X \le 1, \frac{1}{2} < Y \le 1 \right).$$

Solution. Using the joint PDF directly,

$$\begin{aligned} P\left(\frac{1}{2} < X \le 1, \frac{1}{2} < Y \le 1 \right) &= \int_{1/2}^1 \int_{1/2}^1 (x+y)\, dy\, dx \\ &= \int_{1/2}^1 \left[xy + \frac{y^2}{2} \right]_{1/2}^1 dx \\ &= \int_{1/2}^1 \left[\left(x + \frac{1}{2} \right) - \left(\frac{x}{2} + \frac{1}{8} \right) \right] dx \\ &= \int_{1/2}^1 \left(\frac{x}{2} + \frac{3}{8} \right) dx \\ &= \left[\frac{x^2}{4} + \frac{3x}{8} \right]_{1/2}^1 \\ &= \left(\frac{1}{4} + \frac{3}{8} \right) - \left(\frac{1}{16} + \frac{3}{16} \right) \\ &= \frac{5}{8} - \frac{1}{4} \\ &= \frac{3}{8} \end{aligned}$$

(e) (2 marks) Are X and Y independent? Justify your answer.

Solution. Independence would require

$$f_{X,Y}(x,y) = f_X(x) f_Y(y).$$

Here, for $0 \le x, y \le 1$,

$$f_X(x) f_Y(y) = \left(x + \frac{1}{2} \right) \left(y + \frac{1}{2} \right),$$

which is not equal to $x + y = f_{X,Y}(x,y)$ in general. Therefore

$$X \text{ and } Y \text{ are not independent.}$$